

RoSIP: A Scale for Measuring Appearance-Based Social Interaction Potential in Robots

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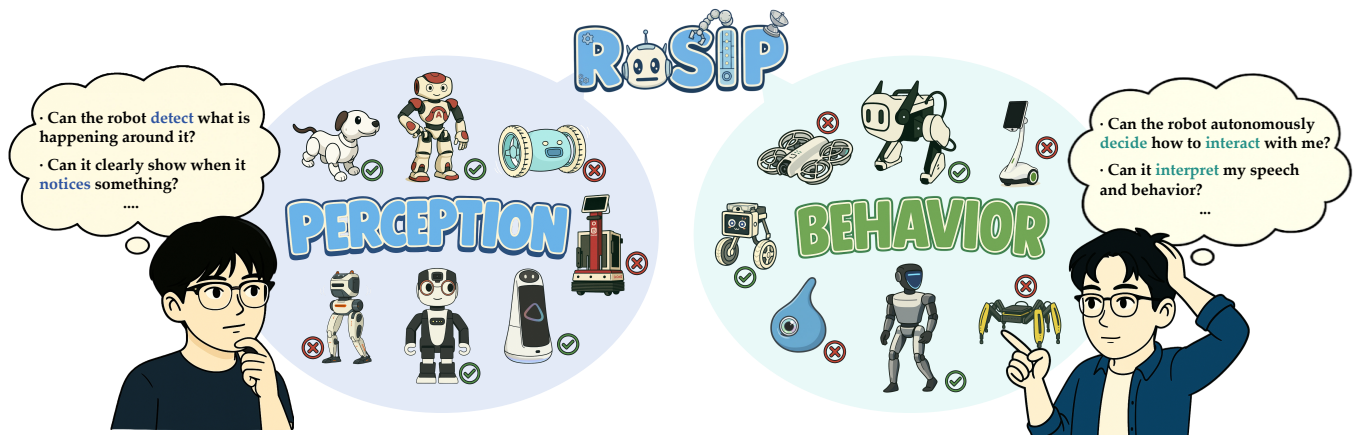


Figure 1: RoSIP measures two appearance-based dimensions of social interaction potential: Perceptual Potential (left) and Behavioral Potential (right). Example robot morphologies illustrate how visual features support or undermine each inference.

Abstract

First impressions are critical for public-facing social robots: users rapidly infer a robot's potential for social interaction from its appearance, shaping expectations and willingness to engage. Yet no existing scale captures how people interpret the interaction potential implied by a robot's visual affordances. We introduce the Robot Social Interaction Potential Scale (RoSIP), a concise appearance-based scale assessing two dimensions—Perceptual Potential and Behavioral Potential. Across a pilot study and large-scale exploratory and confirmatory factor analyses ($N = 750$), we identified a 10-item, two-factor structure with strong internal consistency and solid construct and discriminant validity. RoSIP provides a dedicated tool for rapidly quantifying appearance-based inferences about a robot's social interaction potential, enabling future work to systematically link robot morphology and social perception in HRI.

CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; • **Applied computing** → Psychology.

Keywords

Robot appearance, Social interaction potential, Social Affordances, Social Robots

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1 INTRODUCTION

Robots deployed in public and service environments often interact with users only briefly [19]. In such one-shot encounters, people rapidly form impressions of whether a robot appears able to perceive, understand, or respond to them before any interaction occurs. These appearance-based judgments strongly shape expectations of a robot's interaction potential [2, 28, 30]. For instance, delivery robots lacking eye-like features can make it difficult for



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people to infer orientation or attentional focus, reducing approach willingness [22]. Such features are commonly described as *social affordances*—visual features that communicate potential for social interaction [4, 9, 13, 34].

Despite the importance of appearance-based impressions in Human–Robot Interaction (HRI), there is currently no tool designed to evaluate this construct. Existing scales target broader dimensions—such as anthropomorphism, animacy, mind perception, or social presence—which capture global impressions or mental-state attributions [3, 16, 18, 29]. These constructs, however, are conceptually distinct from users’ perception of the interaction potential conveyed by a robot’s visual affordances. For instance, a robot may still be judged as highly anthropomorphic overall even when it lacks clear features such as face or eye-like structure [40]. In such cases, however, users are less likely to attribute adequate social interaction potential to the robot—particularly perceptual potential, such as the action possibility of flexibly attending to others [36].

To address this gap, we introduce the *Robot Social Interaction Potential Scale (RoSIP)*, an appearance-based measure that captures how users infer a robot’s potential for social interaction solely from its visual affordances. Our focus is on robots intended for social or service-oriented interactions—such as guiding, assisting, or engaging in brief dialogue—where appearance strongly shapes first impressions [9, 13]. This stands in contrast to industrial robots, for which social affordances are not required [33].

2 PREVIOUS WORK

2.1 Social Affordances Behind Human-like Appearance

A large body of work shows that a robot’s human-like appearance reliably shapes social judgments, including trustworthiness [10, 31], likability [24], perceived intelligence [7], and even willingness to interact [11, 23]. These effects have traditionally been explained through high-level theoretical accounts. The mere appearance hypothesis proposes that observers generalize expectations from human–human interaction based on a robot’s overall similarity to the human form [43]. In contrast, mind perception theory argues that people infer mental capacities—such as emotional experience or planning—from specific visual features [16], thereby increasing expectations of social interaction.

However, growing evidence suggests that these appearance-driven effects may not stem solely from human-likeness itself, but may also be influenced by the social affordances implied by specific morphological features. For instance, eye-like features can convey perceptual potential—for example, the possibility of sensing the environment or engaging in gaze-related exchanges. In social perception tasks such as perspective-taking, even robots with relatively low overall anthropomorphism can elicit human perspective-taking when they contain clear perceptual affordances, such as a camera-like head [21, 35–37, 42]. Conversely, robots that appear more anthropomorphic overall but lack such features—for example, with blurred faces—show reliably diminished effects [35, 40, 42].

Together, these findings suggest that overall anthropomorphism and mind perception alone are insufficient to account for appearance-based social judgments, and that social affordances behind appearance constitute an additional, under-characterized pathway through which robot morphology shapes social perception in HRI.

2.2 Dimensions of Social Affordances

The notion of *affordances* originates from Gibson [15], who proposed that objects provide directly perceivable possibilities for action. This notion was later adapted in HRI and extended toward *social affordances*, referring to how a robot’s morphology communicates possibilities for social interaction [4, 9, 13, 34].

Despite the widespread use of the concept, the field lacks a systematic framework specifying the dimensions or underlying structure of social affordances, as well as tools for directly measuring how people perceive them. Instead, most empirical studies infer affordances only indirectly—by examining which visual features elicit downstream social perception effects such as gaze following, mimicry, or perspective-taking—effects that are typically triggered only after observers have already perceived some degree of social interaction potential [1, 20, 35, 39, 42, 43].

Therefore, by reviewing HRI studies that use static robot images, we find that social perception effects are consistently elicited when two types of affordances are present:

Perceptual affordances — visual features that signal the potential to detect, attend to, or gather information from the environment (e.g., eye-like structures [1, 14, 39, 42], camera-like heads [21, 35–37], or abstract ocular indicators [8, 41]).

Behavioral affordances — visual features that signal the potential for socially meaningful behavioral responses (e.g., articulated arms [20, 32, 41, 42], or legs and other locomotive limbs [35, 43]).

From a functional perspective, these two affordances are theoretically justified: they reflect the minimal conditions for social interaction. Perceptual affordances signal potential for information input, whereas behavioral affordances signal potential for information output [12, 38].

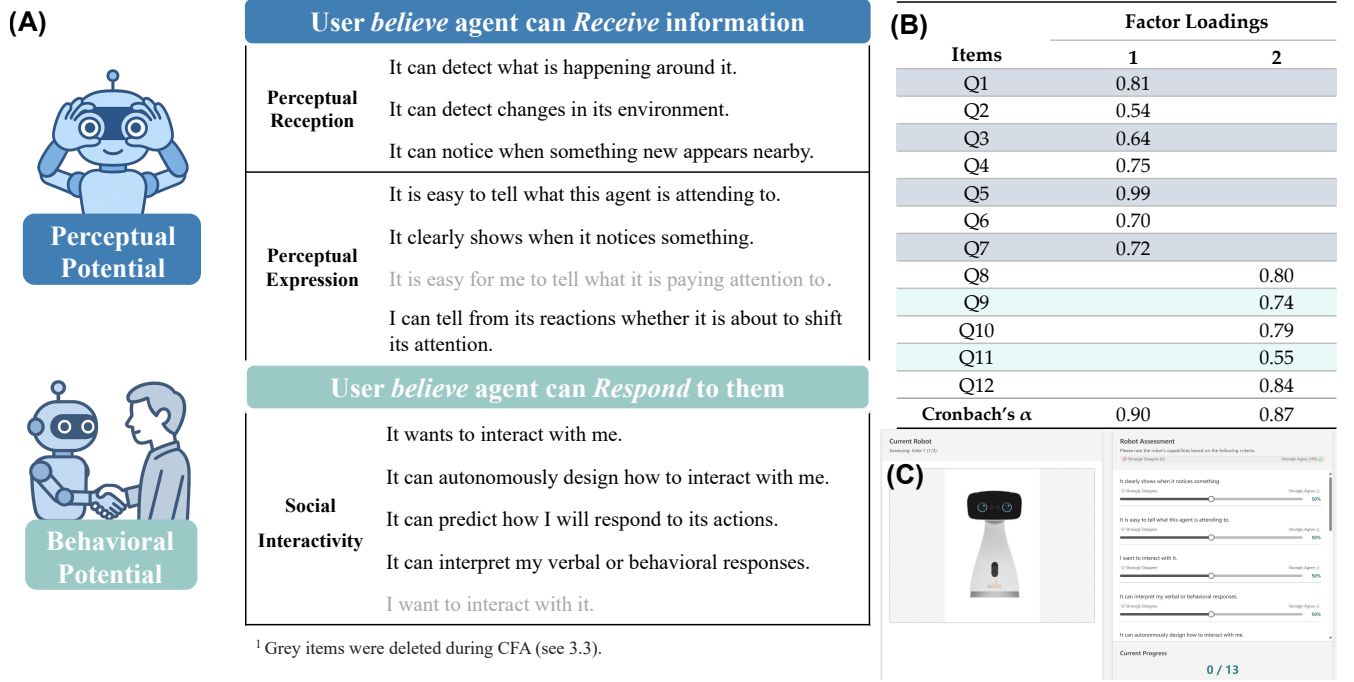
Building on this theoretical distinction and converging empirical evidence, we aim to develop a scale grounded in these two dimensions that directly assesses how users interpret a robot’s social interaction potential from its visual affordances.

3 DEVELOPMENT OF THE ROBOT SOCIAL INTERACTION POTENTIAL SCALE (RoSIP)

3.1 Pilot Study

A pilot study was conducted to assess item clarity and preliminary reliability. The initial item pool drew on established measures related to social perception (e.g., Mind Perception, anthropomorphism, Social Presence, RoSAS) to ensure broad conceptual coverage and appropriate item phrasing [3, 6, 16, 18, 29]. To capture our distinct construct, we additionally incorporated insights from theories of affordances and interpersonal communication [15, 17, 27]. All items were written to reflect two hypothesized dimensions: Perceptual Potential and Behavioral Potential.

Importantly, we use the term *potential* rather than *affordances* to describe what the scale measures. *Social affordances* refer to morphological features that signal possibilities for interaction, while the



¹ Grey items were deleted during CFA (see 3.3).

Figure 2: RoSIP scale development. (A) Items used in EFA. (B) Factor loadings for Perceptual Potential (1) and Behavioral Potential (2) factors with Cronbach's α . (C) Screenshot of the user rating interface.

scale assesses users' inferences about the *social interaction potential* implied by these features. Thus, affordances provide the visual basis, whereas potential reflects the observer's interpretation. Following Norman's distinction, we treat appearance features as *signifiers* that communicate underlying affordances, while RoSIP captures users' inferred social interaction potential rather than affordances per se [26].

We define Perceptual Potential as the user's belief that the agent can receive information from the environment—encompassing perceived abilities to see, detect, or attend to external stimuli (see Figure 2a). This dimension was further divided into two subcomponents: Perceptual Reception, reflecting the belief that the robot can autonomously perceive external stimuli; and Perceptual Expression, reflecting the belief that the robot can outwardly signal that it has perceived something. Behavioral Potential was defined as the user's belief that the agent can generate socially meaningful responses, rather than merely possessing agency or autonomous action-planning capabilities. The initial item pool included 13 items.

Forty-eight participants ($M_{\text{age}} = 21.81$, $SD = 2.06$; 24 males, 24 females) were recruited through university forums and research platforms. The study was conducted online and took approximately 5 minutes to complete. Participants provided informed consent and received 5 RMB (≈ 0.7 USD) as compensation. The study was approved by the Institutional Review Board (IRB) of the authors' institution.

Each participant was randomly assigned one robot image from ABOT, a public dataset of 251 static robot designs [30]. Participants rated all items using a 0–100 slider scale. Items were presented in

the local language (Mandarin Chinese) to ensure comprehension; English translations of all items are provided in the paper, and the full bilingual questionnaire is available at <https://xuconghu.github.io/ABOT2.0/>.

Participants reported no major issues with item comprehension. The scale demonstrated high internal consistency (Perceptual Potential: $\alpha = .90$; Behavioral Potential: $\alpha = .85$), supporting its preliminary reliability. All items were retained, with minor wording refinements made for clarity.

3.2 Factor Extraction

3.2.1 Method. Another 204 participants ($M_{\text{age}} = 23.57$, $SD = 4.58$; 82 males, 120 females, 2 undisclosed) were recruited for the exploratory factor analysis (EFA) through the same recruitment channels as the pilot study and completed the study online. The 13-item preliminary scale retained from the pilot was used. Each participant was randomly assigned to evaluate one robot image from the Abot database (see Figure 2c), with items presented in random order. This one-image-per-participant design was intended to minimize carryover effects while allowing us to sample the full diversity of robot morphologies.

3.2.2 Results. Principal axis factoring with direct oblimin rotation was conducted. One item showing substantial cross-loading and semantic redundancy was removed, resulting in a final 12-item, two-factor solution.

The data were suitable for factor analysis ($KMO = .932$; Bartlett's test: $\chi^2(66) = 1514.71$, $p < .001$). As expected, a two-factor solution emerged, corresponding to Perceptual Potential and Behavioral

Potential. The first two factors had eigenvalues > 1 and jointly explained over 59% of the variance; subsequent factors each explained $< 6\%$. All retained items loaded $\geq .50$ on their respective factors (see Figure 2b). Reliability was high for both subscales (Perceptual Potential $\alpha = .90$; Behavioral Potential $\alpha = .87$). This two-factor structure supports the hypothesized dimensionality of the scale and provides a foundation for subsequent validation.

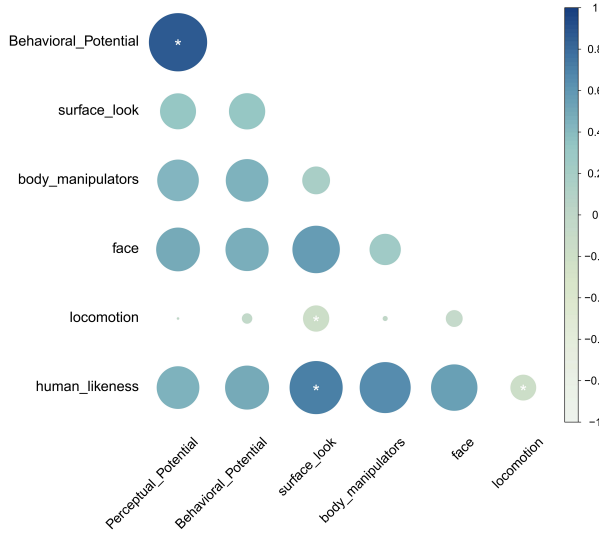


Figure 3: Correlations between the two RoSIP subscales and the human-likeness score and four subscales from Phillips et al [30].

3.3 Validation of Factor Structure

3.3.1 Method. A separate sample of 498 participants ($M_{\text{age}} = 23.61$, $SD = 4.28$; 203 males, 291 females, 4 undisclosed) were recruited through the same recruitment channels for the Confirmatory Factor Analysis (CFA) and completed the study online. Each participant again rated one randomly assigned robot image using the refined scale.

3.3.2 Results. Based on the CFA results, two items with high cross-loadings were removed, yielding a final 10-item scale. These items were also semantically redundant, and their removal did not alter the underlying two-factor structure.

CFA was conducted using robust maximum likelihood estimation. Model fit was evaluated using multiple indices, and the model demonstrated an acceptable fit: $\chi^2(53) = 267.32$, $p < .001$, CFI = .90, TLI = .88, RMSEA = .09 [90% CI: .08, .10], SRMR = .05. All factor loadings were significant, ranging from 0.68 to 0.86, indicating good item convergence.

Regarding reliability, Cronbach's α was .92 for the overall scale, with .90 for Perceptual Potential and .84 for Behavioral Potential, indicating excellent internal consistency across the scale and its two subdimensions.

For convergent validity, the Average Variance Extracted (AVE) and Composite Reliability (CR) values were satisfactory for both subscales, with Perceptual Potential showing AVE = .62, CR = .92,

and Behavioral Potential showing AVE = .54, CR = .86, indicating adequate convergence.

For discriminant validity, the HTMT value between the two subscales was .83, below the recommended .85/.90 thresholds. External correlations with the four anthropomorphism-related subscales (surface look, body manipulators, face, and locomotion) and the overall human-likeness score from Phillips et al. [30] were only moderate ($r = .32-.49$), substantially lower than RoSIP's internal subscale correlation, indicating that RoSIP captures a construct distinct from general anthropomorphism (see Figure 3).

Together, the results validate a stable two-factor structure reflecting the perceptual and behavioral potentials conveyed by the robot's appearance-based affordances.

4 DISCUSSION

This work introduces RoSIP, a concise appearance-based scale of how people infer a robot's social interaction potential. Across factor analysis, the scale demonstrated a stable two-factor structure—Perceptual Potential and Behavioral Potential—with strong reliability and clear discriminant validity from overall anthropomorphism scale. Future work will further establish its distinctiveness through comparisons with other relevant scales. Moreover, the present validation primarily sampled young adults; prior work suggests that age-related differences in attitudes toward robots may shape how appearance-based affordances are interpreted [5, 25]. Future work will extend validation to broader age groups.

Beyond measurement, RoSIP provides a practical lens for robot design: a robot with a camera-like head but no articulated limbs may score high on Perceptual Potential but low on Behavioral Potential, suggesting strong perceptual cues but limited implied interactivity. Designers can then draw on design patterns from robots with higher Behavioral Potential (e.g., simple gestural capabilities) to better align appearance with intended interaction roles.

Going forward, we intend to extend this work in several directions: (1) *Feature-level analysis*. Identifying which specific morphological cues (e.g., eyeball-like structures, camera sensors, articulated limbs) contribute most to each dimension will clarify their functional roles and inform robot design.

(2) *Predictive validity*. A critical next step is to test whether RoSIP scores prospectively predict downstream social-perception effects in HRI (e.g., gaze cueing, joint action). Demonstrating such prospective associations would provide strong evidence for the scale's predictive validity.

(3) *Dynamic realization of affordances*. Although RoSIP captures inferences from static appearance, many affordances are confirmed or violated during subsequent dynamic interaction. Understanding how such matches or mismatches—e.g., robots that appear capable of gesturing but cannot—shape expectations, trust, and engagement will be crucial for real-world deployment.

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